

IBM HR Employee Attrition Analysis

BUSINESS INSIGHTS REPORT

A concise evidence-based view of employee attrition patterns, risk signals, and retention priorities.

Project objective

Identify the workforce segments and job conditions most associated with employee attrition, then translate the analysis into focused, practical retention actions. The project combines exploratory analysis, business queries, and interactive reporting so stakeholders can move from an overall attrition rate to specific risk groups.

Dataset at a glance

1,470

EMPLOYEE RECORDS

35

DATA COLUMNS

Attrition

TARGET VARIABLE

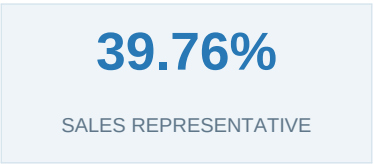
Source: IBM HR Analytics Employee Attrition & Performance. The dataset contains employee demographics, role and compensation details, work conditions, tenure, and an Attrition outcome. Results describe this dataset and should be treated as diagnostic associations, not causal proof.

Methodology and workflow

01	Prepare Review fields, data types, completeness, and analysis-ready variables.
02	Explore Use Python/Jupyter EDA to profile attrition across employee and job dimensions.
03	Query Use MySQL business analysis to validate KPIs and compare priority segments.
04	Communicate Build a Power BI dashboard for interactive filtering and decision support.

Key findings

The strongest risk signals cluster around specific roles and work conditions. Compensation, tenure, travel, and commuting distance also distinguish leavers from employees who stayed.



Overall attrition



Sales department



Sales Representative



Frequent travel



Stock Option Level 0



Distance 21+ km



Overtime gap

Yes	30.53%
No	10.44%

Employee profile comparison

Average	Leavers	Stayers
Monthly income	4,787.09	6,832.74
Tenure (years)	5.13	7.37

Business interpretation

Overtime is the clearest operational warning signal: employees working overtime leave at nearly three times the rate of those who do not. Sales Representatives form the most acute role-based risk group. Lower average income and shorter tenure among leavers suggest that early-career retention and perceived reward may be important intervention points.

Frequent travel, no stock options, and long commutes identify additional groups worth monitoring, especially when multiple risk factors overlap.

Power BI dashboard

The dashboard consolidates headline KPIs and enables users to explore attrition by department, job role, overtime, travel, income, tenure, stock option level, and distance from home.

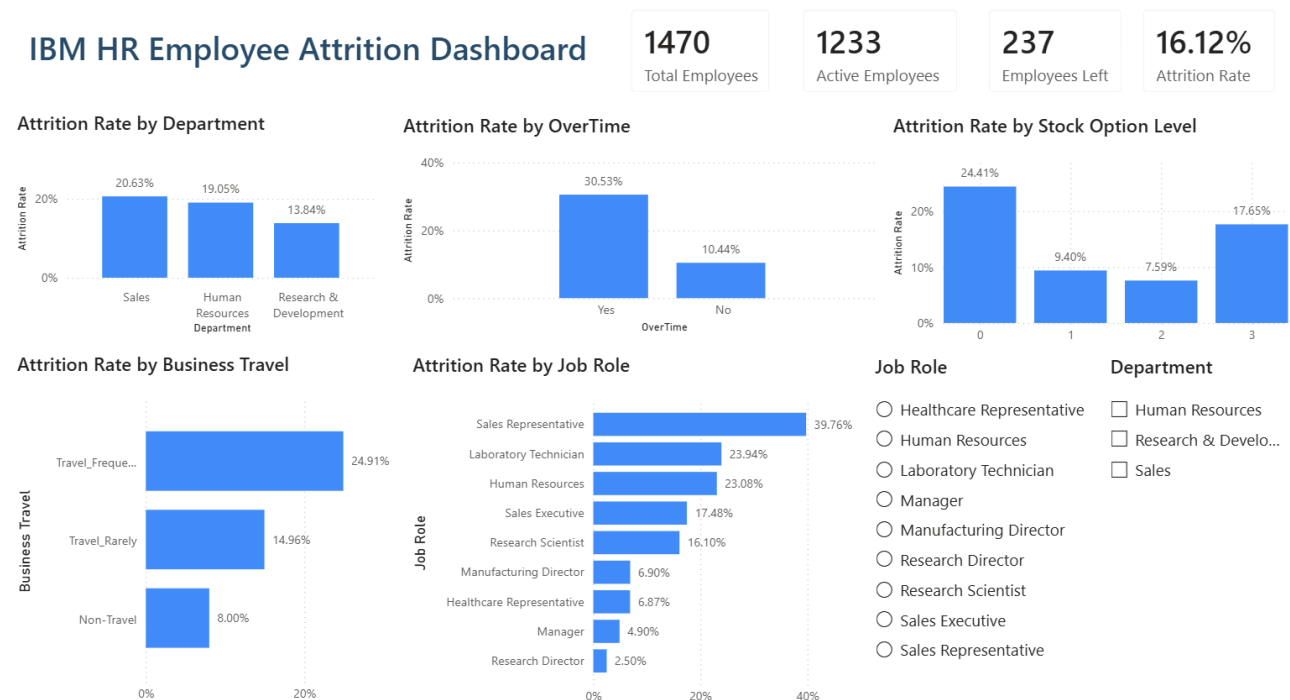


Figure 1. Power BI dashboard summarizing employee attrition KPIs and key risk factors.

Dashboard highlights

237 employees left

The dashboard connects the 16.12% headline rate to actionable workforce segments.

Interactive exploration

Job role and department selectors support focused investigation of attrition patterns.

Recommended business actions

1. Control overtime exposure

Track sustained overtime by team and role, review workload allocation, and trigger manager check-ins before overtime becomes chronic.

2. Prioritize Sales Representative retention

Use role-specific stay interviews, coaching, realistic workload reviews, and clearer progression paths for the highest-risk job group.

3. Strengthen early-tenure support

Introduce structured onboarding checkpoints and manager conversations during the first years, when leavers show shorter average tenure.

4. Review reward and ownership signals

Assess pay positioning and eligibility for stock-based benefits, focusing on lower-income employees and Stock Option Level 0.

5. Reduce travel and commute friction

Offer flexible scheduling or travel rotation where feasible, and monitor frequent travelers and employees commuting 21+ km.

6. Monitor combined risk, not single factors

Use dashboard filters to identify employees or groups exposed to several signals at once, then target support without treating the model as causal.

Conclusion

Attrition in this workforce is concentrated rather than uniform. Overtime and the Sales Representative role are the most prominent signals, while lower income, shorter tenure, frequent travel, no stock options, and longer commutes add meaningful context. The best next step is a focused retention program that reduces avoidable workload pressure, supports early-tenure employees, and tests targeted actions through the Power BI dashboard.